Justin Wong

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PROFESSIONAL EXPERIENCE

GitHub **London, UK**

Software Engineer, Platform Engineering (Notifications) February 2025 – Present

- Helped scale GitHub Notifications from 1.5B to over 8B message deliveries daily, working across product and platform engineering, distributed systems, and site reliability. Used Go, Ruby, MySQL/ Vitess, Redis, Kafka, Kubernetes.
- Acted as availability champion for notifications platform, driving discussions on capacity. Used Datadog telemetry to make system behaviour measurable and validate product, platform and infrastructure changes.
- Rotated on-call to respond to production alerts and service degradation in support of the platform's SLOs.
- Automated the repetitive work, such as building agentic workflows to first-pass postmortems and compile them into weekly SLO reports with concise summaries, saving hours of manual effort weekly.
- Expanded GitHub Notifications from one region to three, migrating 50 asynchronous Kafka-backed jobs across seven workers, three database clusters, and endpoints from a self-hosted data centre to Azure VMs. This involved Kubernetes deployments, VM capacity and system health, and regional cutovers around cross-region dependencies.
- Mitigated impact of traffic growth without adding capacity by improving maintenance-job scheduling with enqueue-once-in-interval, reducing CPU utilisation and database load by 65%.
- Supported live activity push notifications integration on the GitHub mobile app to track GitHub Copilot agent progress, demoed at GitHub Universe 2025 (GitHub's annual developer conference). This included extending our notification integrations for Copilot and building backend implementation to manage live activity mobile tokens.

TradingHub **London, UK**

Software Engineer, Product (Market Abuse and Surveillance) November 2023 – February 2024

- Processed daily batch trading data with quantitative models for market abuse detection (e.g., insider trading, wash trading, spoofing). Used Python, C#, and in-house languages for data scripting and web development.
- Developed data view pagination and order book replay with tick-by-tick playback UI. Restructured the data process with data ossification, eliminating re-computations of same data points; this reduced batch processing times by up to ~30%.

CoreStream **London, UK**

Backend Software Engineer October 2020 – July 2022

- Developed B2B governance, risk, and compliance management software for global clients, incl. Deloitte, Shell, Glencore.
- Owned all facets across the software development lifecycle. Led software design and code sharing sessions.
- Rewrote database queries and triggers to reduce complexity and eliminate deadlocks. Parallelized data computations with C# task-based asynchronous programming, improving site loading times by ~20%. Employed reflection in recursive algorithms on JSON objects. Added Postman and Xunit testing to Azure Pipelines CI/CD to reduce manual testing.

University College London, Department of Computer Science **London, UK**

Full Stack Web Developer July 2020 – October 2020

- Developed a ticketing system with Python Django, used by the UCL CS Department to support online labs. Used Django Channels WebSockets for chat, and Microsoft Graph API for authentication, Microsoft Teams, and calendar integration.

Nuveen (Asset Management, Fixed Income) **London, UK**

Data Engineering Intern May 2019 – September 2019

- Conducted ETLs, from integrating new data sources (Excel spreadsheet and web scraping data collection), to writing scripts for data cleaning, transformations, and core metric reporting and analytics with Python Pandas and Tableau.

EDUCATION

University of Oxford **Oxford, UK**

Master of Science: Software Engineering (Part-Time), Merit October 2022 – May 2026

University College London **London, UK**

Bachelor of Arts and Sciences: Economics Major, Computing Minor. First Class Honors October 2017 – July 2020

EXTRACURRICULARS

2023-2024 CodeYourFuture – Volunteering to help refugees in London learn coding and skills to find technical work.

2024-2025 Phase – Freelance full stack engineering, building a CRM for electricians with AI agentic development.

2024-2025 Outlier – Training generative AI on SDLC tasks, such as code generation accuracy, code review, etc.

2025-2026 Side Projects – Learned OS concepts with projects on implementing containers and ring buffers. Used Rust and C++ to build familiarity with low-level languages, and how standard library APIs interact with OS system calls.